Centro de Investigación y de Estudios Avanzados del Instituto
Politécnico Nacional

DEPARTAMENTO DE FÍSICA

TRABAJO FIN DE GRADO

Sensibilidades a radio de carga de neutrinos en experimentos de neutrinos de reactor involucrando CEνNS

Realizado por
Mario Antonio Guerra Ruiz

Para la obtención del título de

Maestra en Ciencias
en la Especialidad de Física

Dirigido por
Dr. Omar Gustavo Miranda Romagnoli

Ciudad de México, Noviembre de 2026

Centro de Investigación y de Estudios Avanzados del Instituto
Politécnico Nacional

PHYSICS DEPARTMENT

Sensitivities to neutrino charge radius at reactor neutrino experiments involving CEνNS

Presented by
Mario Antonio Guerra Ruiz

In order to obtain the

Master of Science degree
speciality in Physics

Under the supervision of
PhD Omar Gustavo Miranda Romagnoli

Mexico City, November of 2026

Agradecimientos

agradecimiento aqui.

Resumen

resumen aqui

Abstract

resumen en ingles

Contents

1. Introduction

intro

2. Electroweak theory in the Standard Model of particles

2.1. Introduction

The Standard Model of particles (SM) is the most currently utilized theory of interactions of subatomic particles. It describes all matter particles and forces between them, excluding gravity, as interactions between 12 fermions (6 quarks and 6 leptons) with their corresponding anti-fermions, 4 spin-1 force-carrying gauge bosons and 1 scalar boson as shown in Fig. 2.1. Within this framework exist 3 distinct forces: strong nuclear force, mediated by the gluon (g) boson; electromagnetic (EM) force, mediated by the photon (γ) boson; and weak nuclear force, mediated by the $W^{\pm}$ and Z bosons. This work pertains to the description of the properties of the only family of particles that present no electric or color charge known as neutrinos (ν), wherein the SM exist three known flavors (ν_e, ν_μ and ν_τ) corresponding to the latter half of (electro)weak isospin doublets formed with the known charged leptons (e^-, μ^- and τ^-). This chapter aims to describe neutrinos in the SM, the weak force and its origin as a part of a larger broken symmetry known as "Electroweak Theory" (EW).

2.2. Symmetries and fermion content

2.2.1 Neutrinos

Neutrinos are charge-less, color-less, spin-$\frac{1}{2}$ particles with negligible masses compared to other fundamental particles first proposed in 1930 by Pauli W. [1], and later detected by C. L. Cowan et al. in 1956 [2], to solve known apparent contradictions in conserved quantities of beta decays. Beta decay was known to be as the emission of beta particles (electrons) by the nuclei of unstable atoms, wherein the difference in mass of the original and resulting nuclei of the decay should fix the energy at which the emitted particles are detected. Also of note, due to conservation of angular momentum the sum of the spin (S_e) of the electron, the spin of the resulting nucleus (S_{N^*}) and the angular momentum between them (L) should equal the original nucleus' spin (S_N). Despite this, it was know experimentally that the resulting electrons where emitted in a spectrum of energies less than expected and at $L = 0$, making the sum of spins impossible to match. Pauli's solution to the apparent violation of the conservation of these quantities was to hypothesize the existence of a second neutral spin-$\frac{1}{2}$ with very small mass compared to the electron that allows for both the distribution of energy between both emitted particle, thus letting the electron be emitted at different energies depending on the kinematics, and the conservation of

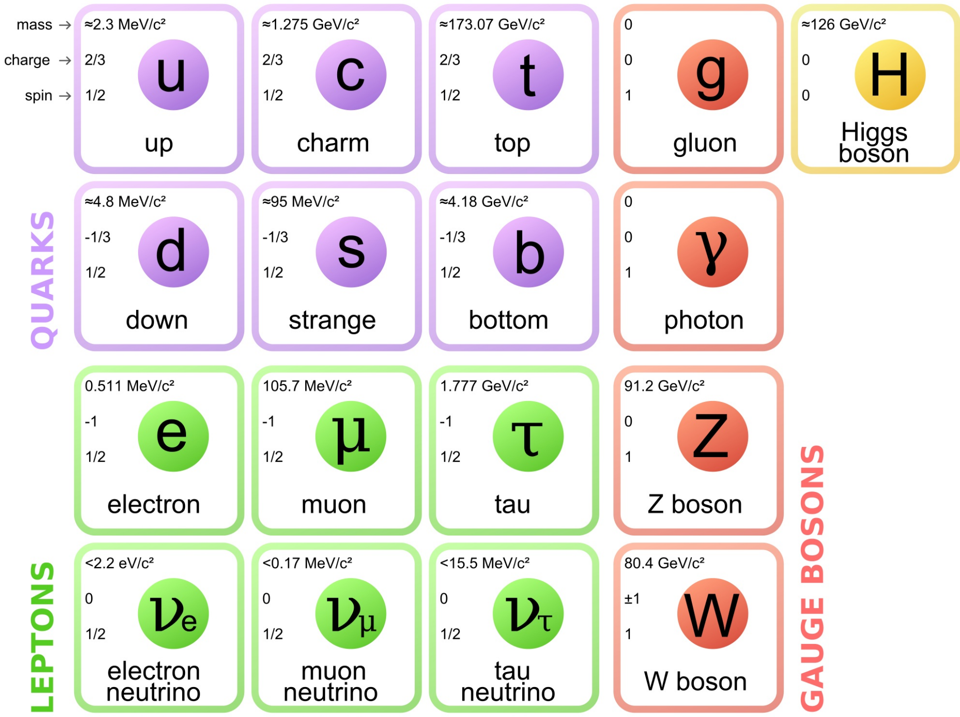


Figure 2.1: *Particles in the SM*

angular momentum.

$$\begin{array}{ccc}
N \to N^* + e^- & \to & N \to N^* + e^- + \nu \\
\hline
m_N = m_{N^*} + E_{e^-} & \to & m_N = m_{N^*} + E_{e^-} + E_\nu \\
S_{z\ N} = S_{z\ N^*} + S_{z\ e^-} & \to & S_{z\ N} = S_{z\ N^*} + S_{z\ e^-} + S_{z\ \nu}
\end{array}$$

Later refinement of the atomic model found that nuclei where made up of positively charged elements named protons (p) and neutrally charged elements known as neutrons[1] (n), leaving the small hypothesized particle to be named neutrino (ν) later by Enrico Fermi. It is well understood now that the beta decay is a process in which a neutron within an unstable nucleus decays into a proton, an electron and a(n) (anti)neutrino.

$$n \to p + e^- + \bar{\nu}$$

Subsequent experiments exploiting cross symmetry of the above interaction [2], accelerator-produced pion decay [3], and charged-current tau neutrino interactions in nuclear emulsion [4, 5] confirmed the existence of three distinct active neutrino flavors and the conservation of lepton family number in the interactions that involved these particles. Given that neutrinos don't interact via electromagnetic or strong forces a theory of a different force that mediated these interactions was developed into what is now known as the "weak nuclear force".

2.2.2 Weak Nuclear Force

Developments in quantum mechanics (QM) and special relativity (SR) led up to the first theories that could describe the behavior of some subatomic particles and their

[1] this was the name first proposed for the particle hypothesized by Pauli, later changed

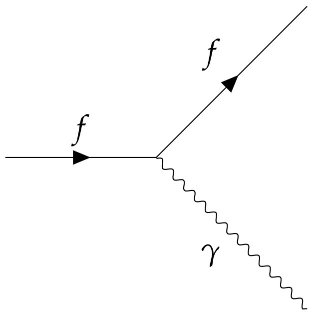
interactions through excitations of quantized fields called quantum field theories (QFTs) obtained by promoting classical fields to operator-valued quantities satisfying canonical (anti)commutation relations and deriving their dynamics from a Lorentz-invariant action principle. One of the earliest and most successful QFTs was Quantum Electro Dynamics (QED), describing how spin-1/2 (anti)particles ($\overset{(-)}{\psi}$) interacted with an EM 4-potential (A^μ) by minimizing the action given by the Lagrangian density:

$$\mathcal{L}_{\text{QED}} = \underbrace{\bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi}_{\text{free Dirac field}} - \underbrace{\frac{1}{4}F_{\mu\nu}F^{\mu\nu}}_{\text{free EM field}} - \underbrace{e\bar{\psi}\gamma^\mu\psi A_\mu}_{\text{interaction}}. \tag{2.1}$$

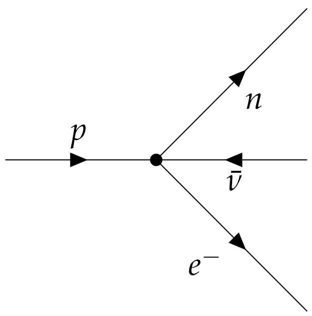
where

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \tag{2.2}$$

is the electromagnetic field strength tensor, ψ is the charged fermion field of mass m, and $D_\mu = \partial_\mu + ieA_\mu$ is the covariant derivative encoding the minimal coupling between the fermion and the gauge field A_μ. While the 'free' terms in (2.1) describe the propagation of Dirac fermions and photons in free Minkowski space, the 'interaction' term describes a vertex that couples the (anti)fermion current with the electromagnetic four-potential, representing the elementary process of a(n) (anti)fermion emitting or absorbing a photon. At zeroth-order $\hbar$ perturbative terms, QED had shown predictive power for processes like Compton ($\gamma + e^- \to \gamma + e^-$) [6] and Møller ($e^- + e^- \to e^- + e^-$) [7] scattering by the time Fermi first proposed an interaction inspired by the QED interaction term, where instead of a fermion current emits a photon, two fermion currents meet at a vertex such that a coupling constant (now called Fermi's constant G_F) quantified the intensity of the interaction, and thus can be estimated experimentally.



$$\sim \quad e\bar{\psi}_f\gamma^\mu\psi_f A_\mu$$



$$\sim \quad \frac{G_F}{\sqrt{2}}(\bar{\psi}_n\gamma^\mu\psi_p)(\bar{\psi}_\nu\gamma_\mu\psi_{e^-})$$

Subsequent discoveries in particle physics revealed that charged-current weak interactions violate parity symmetry maximally. First theorized by Lee and Yang in 1956 [8] to resolve the then-known $\tau - \theta$ puzzle, and later verified experimentally by Wu et al. in 1957 [9], this motivated a modification of the Fermi interaction in the form of a combination of the two bilinear Lorentz-invariant structures that admit such violation [10, 11].

$$\mathrm{V}: j_V^\mu = \bar{\psi}\gamma^\mu\psi \quad \xrightarrow{\mathcal{P}} \quad j'_{V\,\mu} = \bar{\psi}\gamma_\mu\psi \tag{2.3}$$

$$\mathrm{A}: j_A^\mu = \bar{\psi}\gamma^\mu\gamma^5\psi \quad \xrightarrow{\mathcal{P}} \quad j'_{A\,\mu} = -\bar{\psi}\gamma_\mu\gamma^5\psi \tag{2.4}$$

The expressions (2.3) and (2.4) show how different bilinear structures change under parity ($\mathcal{P}$) and how pure vector (axial) currents are $\mathcal{P}-even$ ($\mathcal{P}-odd$). Given a general current structure, $j^\mu = g_V j_V^\mu + g_A j_A^\mu$, observables that depend on the inner product of two such currents transform under parity as:

$$|\mathcal{M}|^2 = (j_1 \cdot j_2) = g_V^2(j_{1\,V} \cdot j_{2\,V}) + g_A^2(j_{1\,A} \cdot j_{2\,A}) + g_V g_A(j_{1\,V} \cdot j_{2\,A} + j_{1\,A} \cdot j_{2\,V}) \tag{2.5}$$

$$\xrightarrow{\mathcal{P}}$$

$$|\mathcal{M}'|^2 = (j'_1 \cdot j'_2) = g_V^2(j'_{1\,V} \cdot j'_{2\,V}) + g_A^2(j'_{1\,A} \cdot j'_{2\,A}) - g_V g_A(j'_{1\,V} \cdot j'_{2\,A} + j'_{1\,A} \cdot j'_{2\,V}). \tag{2.6}$$

The only difference between (2.5) and (2.6) lies in the sign of the interference term, such that the degree of $\mathcal{P}$-symmetry violation is given by the coefficient $\frac{g_V g_A}{g_V^2 + g_A^2}$. Normalizing this there are two options for maximally violated $\mathcal{P}$-symmetry given by $\frac{g_V}{g_A} = \pm 1$, also known as $V \pm A$ currents. Given that in the Wu experiment it was determined that (anti)neutrinos are emitted exclusively as left(right)-handed fermions the appropriate structure to use for weak interactions was determined to be $V - A$. Modifying the Fermi Lagrangian density for beta decay to include this gives the following expression:

$$\mathcal{L}_{CC} = \frac{G_F}{\sqrt{2}} \left[\bar{\psi}_{\nu_e}\gamma^\mu(1-\gamma^5)\psi_e\right] \left[\bar{\psi}_p\gamma_\mu(1-\gamma^5)\psi_n\right] + \text{h.c.} \tag{2.7}$$

Later refinements of this theory succeeded in demonstrating the universality of the weak interaction across leptonic decays and strangeness-changing meson decays through Cabibbo mixing [12]. This not only established that all such processes are mediated by the same underlying force, but also motivated the prediction and subsequent discovery of the charm quark [13]. With these improvements on the theory thus far, the point-like structure of the interaction still led to scattering amplitudes proportional to the center-of-mass energy $\sqrt{s}$, resulting in S-matrix unitarity violation at higher energies. The natural resolution, in analogy with QED, is to conceive of Fermi's interaction as the low-energy limit of a gauge field theory, wherein the maximal parity violation of the charged currents naturally gives rise to three mediating vector bosons corresponding to the generators of an $SU(2)_L$ symmetry, which must be massive due to the short range of the weak interaction. Given this hypothesis, Glashow showed in 1961 [14] that the gauge field theory whose low-energy limit reproduces the $V - A$ structure of charged currents requires a $SU(2)_L \times U(1)_Y$ symmetry group, which not only predicted the existence of neutral weak currents, later measured in 1973 at the GARGAMELLE experiment at

CERN [15, 16], but also unified the weak and electromagnetic interactions into a single electroweak theory, later completed independently by Weinberg and Salam through the incorporation of spontaneous symmetry breaking [17, 18].

2.3. Massive and Massless Mediators

2.3.1 Photons

It is well established within QED that electromagnetic interactions between charged fermions are mediated by a bosonic gauge field. The mediating boson, the photon, is massless as required by $U(1)$ gauge invariance, a property that manifests physically as the infinite range of the electromagnetic interaction, and carries spin-1. Its dynamics are governed by (2.1), and by applying the Euler-Lagrange equations to the free electromagnetic field term with respect to A^μ, one obtains the equations of motion:

$$\partial_\mu \left(\frac{\partial \mathcal{L}_{\text{free EM}}}{\partial(\partial_\mu A_\nu)} \right) - \frac{\partial \mathcal{L}_{\text{free EM}}}{\partial A_\nu} = 0 \implies \partial_\mu F^{\mu\nu} = \partial^2 A^\nu - \partial^\nu(\partial_\mu A^\mu) = 0, \tag{2.8}$$

which in momentum space, with $\partial_\mu \to -iq_\mu$, becomes

$$\left(-q^2 \eta^{\mu\nu} + q^\mu q^\nu \right) \widetilde{A}_\nu = K^{\mu\nu} \widetilde{A}_\nu = 0. \tag{2.9}$$

The propagator $D^{\mu\nu}$, defined by $K^{\mu\alpha} D_{\alpha\nu} = \delta^\mu{}_\nu$, cannot be obtained directly from the kinetic operator $K^{\mu\nu}$, since the latter is singular by construction: the action is invariant under $U(1)$ gauge transformations $A^\mu \to A'^\mu = A^\mu + \partial^\mu \lambda$, which implies that $K^{\mu\nu}$ has a zero eigenvalue along the gauge orbit. A propagator is only obtainable upon gauge fixing, which is implemented by augmenting the Lagrangian density with a gauge-fixing term,

$$\mathcal{L}'_{\text{free EM}} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2\xi} \left(\partial_\mu A^\mu \right)^2, \tag{2.10}$$

such that the modified kinetic operator

$$K'^{\mu\nu} = -q^2 \eta^{\mu\nu} + \left(1 - \frac{1}{\xi} \right) q^\mu q^\nu \tag{2.11}$$

is now invertible for any $\xi \neq 0$, yielding the generalized R_ξ propagator

$$D_\gamma^{\mu\nu}(q) = \frac{-i}{q^2 + i\epsilon} \left[\eta^{\mu\nu} - (1 - \xi) \frac{q^\mu q^\nu}{q^2} \right], \tag{2.12}$$

where any choice of $\xi \in \mathbb{R} \setminus \{0\}$ yields equivalent physical predictions, as gauge-dependent terms cancel in all observable quantities.

2.3.2 $W^\pm$ and Z^0

In contrast to the photon, the weak interaction's short range is a consequence of its mediators' mass and thus its dynamics are no longer governed by a Maxwell Lagrangian density, but rather one that includes a mass term that breaks the $U(1)$

found in the original, called a Proca Lagrangian, with a propagator that no longer requires gauge fixing but still is invariant under Lorentz transformations.

$$\mathcal{L}_{\text{Proca}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}M^2 A_\mu A^\mu. \tag{2.13}$$

In calculating its propagator by solving the equations of motion and inverting the kinetic operator in momentum space one finds that:

$$\partial_\mu \left(\frac{\partial \mathcal{L}_{\text{Proca}}}{\partial(\partial_\mu A_\nu)}\right) - \frac{\partial \mathcal{L}_{\text{Proca}}}{\partial A_\nu} = 0 \implies \partial_\mu F^{\mu\nu} + M^2 A^\nu = \partial^2 A^\nu - \partial^\nu(\partial_\mu A^\mu) + M^2 A^\nu = 0, \tag{2.14}$$

and by taking the four-divergence ∂_ν of (2.14),

$$\begin{aligned} \partial_\nu \partial^2 A^\nu - \partial_\nu \partial^\nu(\partial_\mu A^\mu) + M^2(\partial_\mu A^\nu) = 0, \\ \implies M^2(\partial_\mu A^\nu) = 0. \end{aligned} \tag{2.15}$$

Now taking $M^2 \neq 0$ the gauge given by $\partial_\mu A^\mu = 0$ is no longer a choice but rather an equation of motion that reduces the four degrees of freedom of the components of A^μ to two transverse and one longitudinal polarization modes of a massive spin-1 particle. Writing the equations of motion in momentum space with $\partial^\mu \to -iq^\mu$:

$$(-q^2\eta^{\mu\nu} + q^\nu q_\mu + M^2)A_\mu = ((M^2 - q^2)\eta^{\mu\nu} + q^\nu q_\mu)A_\mu = 0. \tag{2.16}$$

Inverting $K^{\mu\nu}_{\text{Proca}} = ((M^2 - q^2)\eta^{\mu\nu} + q^\mu q^\nu)$ gives a unique choice for the Proca propagator $D^{\mu\nu}_{\text{Proca}}$ such that $K^{\mu\alpha}_{\text{Proca}} D_{\text{Proca}\,\alpha\nu} = \delta^\mu{}_\nu$.

$$D^{\mu\nu}_{\text{Proca}} = \frac{i}{q^2 - M^2 + i\epsilon}\left[\eta^{\mu\nu} - \frac{q^\mu q^\nu}{M^2}\right]. \tag{2.17}$$

It is worth noting that interactions mediated by (2.17) between conserved currents, for which $\partial_\mu j^\mu = 0$ implies $q_\mu \tilde{j}^\mu = 0$, reduce to the Fermi effective form in the limit $|q^2| \ll M^2$. Expanding the denominator,

$$\frac{1}{q^2 - M^2} = \frac{-1}{M^2}\left[\frac{1}{1 - q^2/M^2}\right] \xrightarrow{|q^2| \ll M^2} \frac{-1}{M^2}\left[1 + \frac{q^2}{M^2} + \mathcal{O}\left(\frac{q^4}{M^4}\right)\right], \tag{2.18}$$

where the amplitude of an interaction is now

$$i\mathcal{M} = (-ig)^2\, j^\dagger_\mu\, D^{\mu\nu}_{\text{Proca}}\, j_\nu \xrightarrow{|q^2| \ll M^2} \frac{ig^2}{M^2}\, j^\dagger_\mu\, \eta^{\mu\nu}\, j_\nu, \tag{2.19}$$

which upon comparison with the Fermi four-fermion amplitude $i\mathcal{M}_{\text{Fermi}} = i\frac{G_F}{\sqrt{2}}\, j^\dagger_\mu\, \eta^{\mu\nu}\, j_\nu$ yields the tree-level matching condition

$$\frac{G_F}{\sqrt{2}} \sim \frac{g^2}{M_W^2}, \tag{2.20}$$

where its actual matching relation will be given when calculating the $SU(2)_L$ couplings with $W^\pm$ and charged currents.

2.4. Electroweak mixing and $\sin^2\theta_W$

2.4.1 $SU(2)_L$

As established in the previous section, the inclusion of massive spin-1 bosons explicitly breaks the usual $U(1)$ symmetry found in QED and thus a gauge theory that accommodates such mediators requires a larger symmetry group. A natural candidate comes in the study of a theory with an $SU(2)$ symmetry whose three generators,

$$\tau^1 1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \tau^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \tau^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{2.21}$$

couple exclusively with left(right)-handed (anti)fermions (as is the case for charged current weak interactions) that act on doublets of the form

$$L = \begin{pmatrix} \nu_l \\ l \end{pmatrix}_L, \tag{2.22}$$

where l denotes any charged lepton and ν_l it's corresponding (anti)neutrino. A theory of fermion interactions with a local $SU(2)_L$ symmetry requires promoting the partial derivative in the Dirac Lagrangian to a covariant derivative

$$D_\mu = \partial_\mu + ig\frac{\tau^a}{2}W^a_\mu = \partial_\mu + ig\left(\frac{\tau^1}{2}W^1_\mu + \frac{\tau^2}{2}W^2_\mu + \frac{\tau^3}{2}W^3_\mu\right), \tag{2.23}$$

and introducing a non-abelian field strength tensor for each gauge boson W^a, $a \in \{1, 2, 3\}$,

$$W^a_{\mu\nu} = \partial_\mu W^a_\nu - \partial_\nu W^a_\mu - g\epsilon^{abc}W^b_\mu W^c_\nu, \tag{2.24}$$

where the last term, absent in the abelian case of QED, encodes the self-interactions among the gauge bosons that arise from the non-abelian nature of $SU(2)_L$. The resulting Lagrangian density is

$$\mathcal{L}_{SU(2)_L} = \bar{L}\, i\gamma^\mu D_\mu L - \frac{1}{4}W^a_{\mu\nu}W^{a\mu\nu}, \tag{2.25}$$

where $L = \binom{\nu_l}{l}_L$ denotes the left-handed lepton doublet. Extracting and expanding the interaction term in (2.25) explicitly gives,

$$\begin{aligned} \mathcal{L}_{SU(2)_L,\,\text{int}} &= -\frac{g}{2}\bar{L}\gamma^\mu(\tau^1 W^1_\mu + \tau^2 W^2_\mu + \tau^3 W^3_\mu)L \\ &= -\frac{g}{2}\begin{pmatrix} \bar{\nu}_l & \bar{l} \end{pmatrix}_L \gamma^\mu \begin{pmatrix} W^3_\mu & W^1_\mu - iW^2_\mu \\ W^1_\mu + iW^2_\mu & -W^3_\mu \end{pmatrix} \begin{pmatrix} \nu_l \\ l \end{pmatrix}_L. \end{aligned} \tag{2.26}$$

Identifying that the upper right and lower left entries of the matrix in (**??**) correspond neatly to the known charged current weak interactions makes the definition of the $W^\pm$ bosons such that $W^\pm_\mu = \frac{1}{\sqrt{2}}(W^1_\mu \mp iW^2_\mu)$ a way to separate the interaction Lagrangian density into a charged current (CC) term and a neutral current (NC) term.

$$\mathcal{L}_{\text{int}} = \underbrace{-\frac{g}{2}\left[\bar{\nu}_L\gamma^\mu\nu_L - \bar{l}_L\gamma^\mu l_L\right]W^3_\mu}_{\mathcal{L}_{\text{NC int}}} \underbrace{-\frac{g}{\sqrt{2}}\left[\bar{\nu}_L\gamma^\mu l_L\, W^+_\mu + \bar{l}_L\gamma^\mu\nu_L\, W^-_\mu\right]}_{\mathcal{L}_{\text{CC int}}}. \tag{2.27}$$

It is worth noting that the amplitude of a CC interaction computed from $\mathcal{L}_{CC\,\text{int}}$ in (2.27) in the limit $|q^2| \ll M_W^2$ reproduces the Fermi four-fermion amplitude, yielding the tree-level matching relation between G_F and g.

$$
\begin{aligned}
-i\mathcal{M} &= \left[\frac{-ig}{\sqrt{2}}\bar{\nu}_L\gamma_\mu l_L\right]\frac{-i\eta^{\mu\nu}}{M_W^2-q^2}\left[\frac{-ig}{\sqrt{2}}\bar{l}_L\gamma_\nu\nu_L\right] && (2.28)\\
&= \left[\frac{-ig}{\sqrt{2}}\bar{\nu}\frac{1}{2}\gamma_\mu(1-\gamma^5)l\right]\frac{-i\eta^{\mu\nu}}{M_W^2-q^2}\left[\frac{-ig}{\sqrt{2}}\bar{l}\gamma_\nu(1-\gamma^5)\nu\right] \\
&\overset{|q^2|\ll M_W^2}{\approx} \frac{ig^2}{8M_W^2}\left[\bar{\nu}\gamma^\mu(1-\gamma^5)l\right]\left[\bar{l}\gamma^\mu(1-\gamma^5)\nu\right]. && (2.29)
\end{aligned}
$$

Comparing (2.29) with the amplitude given by (2.7) yields the relation

$$
\frac{G_F}{\sqrt{2}} = \frac{g^2}{8M_W^2}, \tag{2.30}
$$

which is of particular importance as it expresses the experimentally well-measured Fermi constant directly in terms of the $SU(2)_L$ gauge coupling and the $W^\pm$ mass, thereby connecting the low-energy effective theory to the underlying electroweak gauge structure and providing a precise prediction for M_W once g and G_F are independently determined. Given this and the knowledge that weak neutral currents exist, it is tempting to identify W_μ^3 as their mediator. However, this cannot be the case since this term couples exclusively to left(right)-handed (anti)fermions, while it is experimentally well established that weak neutral current interactions do not maximally violate $\mathcal{P}$-symmetry as charged current interactions do [15, 16]. The $SU(2)_L$ symmetry is therefore too chiral to accommodate weak neutral currents, which while not $\mathcal{P}$-conserving, do not exhibit the maximal parity violation of the charged current sector. This incompatibility however points to the possibility of a more complete theory wherein the purely chiral $SU(2)_L$ generators are combined with a $\mathcal{P}$-conserving $U(1)_Y$ hypercharge gauge field, analogous to that of QED, to model the weak and electromagnetic interactions as a single unified force known as the Electroweak (EW) theory.

2.4.2 $SU(2)_L \times U(1)_Y$

To establish a theory coherent with the known properties of leptons, it is first necessary to fix the hypercharge (Y) assignments of the left-handed fermion doublet $L = \binom{\nu_l}{l}_L$ and the right-handed lepton singlet $R = (l_R)$ by relating them to the known electric charges Q and weak isospin eigenvalues T^3 via the Gell-Mann–Nishijima relation

$$
Q = T^3 + \frac{Y}{2}. \tag{2.31}
$$

Fixing these gives the values presented in Table 2.1 for each lepton type, noting that right-handed neutrinos are absent from the spectrum due to the lack of experimental evidence for their existence, with the exception of the measured neutrino mass splittings among the three flavors, which falls outside the scope of this work. With this, the full Lagrangian density with the necessary assigned values is:

$$
\mathcal{L}_{EW} = -i\bar{L}\gamma^\mu D_\mu L - i\bar{R}\gamma^\mu D_\mu R - \frac{1}{4}W^{a\,\mu\nu}W^a_{\mu\nu} - \frac{1}{4}B^{\mu\nu}B_{\mu\nu} \tag{2.32}
$$

Multiplet	T^3	Q	Y
$\nu_{l\,L}$	$\frac{1}{2}$	0	-1
l_L	-$\frac{1}{2}$	-1	-1
l_R	0	-1	-2

Table 2.1: *Values for Q, T^3 and Y for each lepton, where l denotes any of the known flavours of charged lepton*

Where $B^{\mu\nu}$ is defined the same way as $F^{\mu\nu}$ in (2.2) for a massless boson field B^μ. The covariant derivative is given by

$$D_\mu = \partial_\mu + ig(\frac{\tau^1}{2}W^1_\mu + \frac{\tau^2}{2}W^2_\mu + \frac{\tau^3}{2}W^3_\mu) + ig'\frac{Y}{2}B_\mu. \tag{2.33}$$

Much like (2.26), the interaction terms of (2.32) can be separated into CC and NC contributions, wherein the NC terms can be re-expressed as a linear combination of two mass eigenstates mixed through an orthogonal rotation parametrized by a single angle known as the Weinberg angle (θ_W). The EW interaction terms are given by

$$\begin{aligned}\mathcal{L}_{EW\,\text{int}} &= -\frac{g}{2}\begin{pmatrix}\bar{\nu}_l & \bar{l}\end{pmatrix}_L \gamma^\mu \begin{pmatrix} W^3_\mu & W^1_\mu - iW^2_\mu \\ W^1_\mu + iW^2_\mu & -W^3_\mu\end{pmatrix}\begin{pmatrix}\nu_l \\ l\end{pmatrix}_L - \frac{g'}{2}\left(\bar{\psi}\gamma^\mu Y\psi\right)B_\mu \\ &= \underbrace{-\frac{g}{\sqrt{2}}\left[\bar{\nu}_L\gamma^\mu l_L\, W^+_\mu + \bar{l}_L\gamma^\mu\nu_L\, W^-_\mu\right]}_{\mathcal{L}_{CC\,\text{int}}}\underbrace{-\frac{g}{2}\left[\bar{\nu}_L\gamma^\mu\nu_L - \bar{l}_L\gamma^\mu l_L\right]W^3_\mu - \frac{g'}{2}\bar{\psi}\gamma^\mu Y B_\mu\psi}_{\mathcal{L}_{NC\,\text{int}}},\end{aligned} \tag{2.34}$$

where ψ denotes fermions with hypercharge Y and weak isospin T^3, the latter being explicitly dependent on the chirality of the field, as left-handed fermions carry $T^3 = \pm 1/2$ while right-handed fermions are $SU(2)_L$ singlets with $T^3 = 0$.

$$\begin{aligned}\mathcal{L}_{NC\,\text{int}} &= -\frac{g}{2}\left[\bar{\nu}_L\gamma^\mu\nu_L - \bar{l}_L\gamma^\mu l_L\right]W^3_\mu - \frac{g'}{2}\bar{\psi}\gamma^\mu Y B_\mu\psi \\ &= -\frac{g}{2}\bar{\psi}\gamma^\mu(2T^3)\psi - \frac{g'}{2}\bar{\psi}\gamma^\mu Y B_\mu\psi \\ &= -\bar{\psi}\gamma^\mu(gT^3W^3_\mu + g'\frac{Y}{2}B_\mu)\psi.\end{aligned} \tag{2.35}$$

The W^3_μ and B_μ fields correspond to the unmixed states that can be combined into the known mass eigenstates A_μ and Z_μ through the orthogonal transformation given by:

$$\begin{pmatrix}B_\mu \\ W^3_\mu\end{pmatrix} = \begin{pmatrix}\cos\theta_W & -\sin\theta_W \\ \sin\theta_W & \cos\theta_W\end{pmatrix}\begin{pmatrix}A_\mu \\ Z_\mu\end{pmatrix}. \tag{2.36}$$

Substituting (2.36) into (2.35),

$$\begin{aligned}\mathcal{L}_{NC\,\text{int}} = &-\left(g\sin\theta_W T^3 + g'\cos\theta_W\frac{Y}{2}\right)\bar{\psi}\gamma^\mu\psi\, A_\mu \\ &-\left(g\cos\theta_W T^3 - g'\sin\theta_W\frac{Y}{2}\right)\bar{\psi}\gamma^\mu\psi\, Z_\mu.\end{aligned} \tag{2.37}$$

Requiring that A_μ corresponds to the photon field that couples with fermions such that $\mathcal{L}_{\text{QED int}} = -e\bar{\psi}\gamma^\mu Q\psi$ where Q follows from (2.31) gives the relations between the coupling constants

$$e = g\sin\theta_W = g'\cos\theta_W. \tag{2.38}$$

Substituting (2.31) and (2.38) into the Z_μ coefficient of (2.37) yields the Z boson interaction term expressed solely in terms of T^3 and Q,

$$\mathcal{L}_{NC\,\text{int}} = -eQ\bar{\psi}\gamma^\mu\psi\, A_\mu - \frac{g}{\cos\theta_W}\left(T^3 - Q\sin^2\theta_W\right)\bar{\psi}\gamma^\mu\psi\, Z_\mu \tag{2.39}$$

3. Radiative Corrections and Neutrino Charge Radius

3.1. Introduction

3.2. Neutrino Charge Radius

3.3. One-loop Corrections to the $\nu\nu\gamma$ vertex

3.4. NCR as a radiative correction of Electroweak Mixing

4. Coherent Elastic Neutrino-Nucleus Scattering

4.1. Introduction

4.2. Cross Section

4.3. Experiments

4.4. Quenching Factors

4.5. Neutrino Sources

4.6. Form Factors

5. SBC-ININ-LAr

5.1. Results

6. Summary and conclusions

Bibliography

[1] Wolfgang Pauli. "Letter to the Physical Institute of the Federal Institute of Technology, Zürich". Open letter to the Gauverein meeting in Tübingen. Reproduced in L. M. Brown, "The idea of the neutrino," *Physics Today*, vol. 31, no. 9, pp. 23–28, 1978. Also in R. Kronig and V. F. Weisskopf (eds.), *Collected Scientific Papers by Wolfgang Pauli*, vol. 2, Interscience, New York, 1964. 1930. URL: http://www.math.utah.edu/~beebe/talks/2015/qtm/pdf/pauli-1930-ltc.pdf.

[2] C. L. Cowan et al. "Detection of the Free Neutrino: a Confirmation". In: *Science* 124.3212 (1956), pp. 103–104. DOI: 10.1126/science.124.3212.103. eprint: https://www.science.org/doi/pdf/10.1126/science.124.3212.103. URL: https://www.science.org/doi/abs/10.1126/science.124.3212.103.

[3] G. Danby et al. "Observation of High-Energy Neutrino Reactions and the Existence of Two Kinds of Neutrinos". In: *Phys. Rev. Lett.* 9 (1 1962), pp. 36–44. DOI: 10.1103/PhysRevLett.9.36. URL: https://link.aps.org/doi/10.1103/PhysRevLett.9.36.

[4] K. Kodama et al. "Observation of tau neutrino interactions". In: *Physics Letters B* 504.3 (Apr. 2001), 218–224. ISSN: 0370-2693. DOI: 10.1016/s0370-2693(01)00307-0. URL: http://dx.doi.org/10.1016/S0370-2693(01)00307-0.

[5] K. Kodama et al. "Final tau-neutrino results from the DONuT experiment". In: *Phys. Rev. D* 78 (5 2008), p. 052002. DOI: 10.1103/PhysRevD.78.052002. URL: https://link.aps.org/doi/10.1103/PhysRevD.78.052002.

[6] O. Klein and Y. Nishina. "Über die Streuung von Strahlung durch freie Elektronen nach der neuen relativistischen Quantendynamik von Dirac". In: *Zeitschrift für Physik* 52.11-12 (Nov. 1929), 853–868. ISSN: 0044-3328. DOI: 10.1007/bf01366453. URL: http://dx.doi.org/10.1007/BF01366453.

[7] C. Møller. "Zur Theorie des Durchgangs schneller Elektronen durch Materie". In: *Annalen der Physik* 406.5 (1932), pp. 531–585. DOI: https://doi.org/10.1002/andp.19324060506. eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/andp.19324060506. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/andp.19324060506.

[8] T. D. Lee and C. N. Yang. "Question of Parity Conservation in Weak Interactions". In: *Phys. Rev.* 104 (1956), pp. 254–258. DOI: 10.1103/PhysRev.104.254.

[9] C. S. Wu et al. "Experimental Test of Parity Conservation in Beta Decay". In: *Phys. Rev.* 105 (1957), pp. 1413–1415. DOI: 10.1103/PhysRev.105.1413.

[10] E. C. G. Sudarshan and R. E. Marshak. "Chirality Invariance and the Universal Fermi Interaction". In: *Phys. Rev.* 109 (1958), pp. 1860–1862. DOI: 10.1103/PhysRev.109.1860.

[11] R. P. Feynman and M. Gell-Mann. "Theory of the Fermi Interaction". In: *Phys. Rev.* 109 (1958), pp. 193–198. DOI: 10.1103/PhysRev.109.193.

[12] N. Cabibbo. “Unitary Symmetry and Leptonic Decays”. In: *Phys. Rev. Lett.* 10 (1963), pp. 531–533. DOI: 10.1103/PhysRevLett.10.531.

[13] S. L. Glashow, J. Iliopoulos, and L. Maiani. “Weak Interactions with Lepton-Hadron Symmetry”. In: *Phys. Rev. D* 2 (1970), pp. 1285–1292. DOI: 10.1103/PhysRevD.2.1285.

[14] Sheldon L. Glashow. “Partial-symmetries of weak interactions”. In: *Nuclear Physics* 22.4 (1961), pp. 579–588. ISSN: 0029-5582. DOI: https://doi.org/10.1016/0029-5582(61)90469-2. URL: https://www.sciencedirect.com/science/article/pii/0029558261904692.

[15] F.J. Hasert et al. “Search for elastic muon-neutrino electron scattering”. In: *Physics Letters B* 46.1 (1973), pp. 121–124. ISSN: 0370-2693. DOI: https://doi.org/10.1016/0370-2693(73)90494-2. URL: https://www.sciencedirect.com/science/article/pii/0370269373904942.

[16] F.J. Hasert et al. “Observation of neutrino-like interactions without muon or electron in the gargamelle neutrino experiment”. In: *Physics Letters B* 46.1 (1973), pp. 138–140. ISSN: 0370-2693. DOI: https://doi.org/10.1016/0370-2693(73)90499-1. URL: https://www.sciencedirect.com/science/article/pii/0370269373904991.

[17] Steven Weinberg. “A Model of Leptons”. In: *Phys. Rev. Lett.* 19 (21 1967), pp. 1264–1266. DOI: 10.1103/PhysRevLett.19.1264. URL: https://link.aps.org/doi/10.1103/PhysRevLett.19.1264.

[18] Abdus Salam and J. C. Ward. “Weak and electromagnetic interactions”. In: *Il Nuovo Cimento* 11.4 (Feb. 1959), 568–577. ISSN: 1827-6121. DOI: 10.1007/bf02726525. URL: http://dx.doi.org/10.1007/BF02726525.